Final Project Overview

Project overview:

Put your knowledge and skills to the test in this hands-on project. Apply everything you've learned throughout the course to complete a data visualization project from start to finish. Connect to a CSV file and import; create interactive visualizations and insightful dashboards while adhering to the best data visualization, storytelling, and dashboard design practices.

Project scenario:

In this scenario, you're a consultant hired by the New York City Tourism Board. The tourism board wants to understand the Airbnb market in the city, particularly focusing on factors like pricing, location, and accommodation types. Your task is to analyze trends, identify potential areas of interest for tourists, and ultimately create an interactive dashboard to share your findings. This will help the tourism board make informed decisions about promoting different areas and types of Airbnb listings to attract visitors.

You searched for the data set that can analyze and explore Airbnb listings in New York City. You found the New York City Airbnb Open data in Kaggle.

You recently completed a course on Tableau from IBM. You thought of putting your newly acquired knowledge and skills to the test by completing this Airbnb data visualization project from start to finish. The project aims to demonstrate your proficiency in Tableau, showcasing your ability to build hierarchies, develop simple data models, and create interactive visualizations using calculations, and interactive dashboards using actions. It is crucial to adhere to best practices in data visualization and dashboard/story design throughout the project. You can share your project through the shareable link to your dashboard to showcase your expertise on Tableau Public with your peers.

About the data set:

Since 2008, guests and hosts have used Airbnb to expand on traveling possibilities and present a more unique, personalized way of experiencing the world. This data set describes the listing activity and metrics in New York city for 2019. This data file includes all the needed information to find out more about hosts, geographical availability, and necessary metrics to make predictions and draw conclusions. This public data set is part of Airbnb.

This public data set is available on the Kaggle website as *New York City Airbnb Open Data* under the *CC0: Public Domain* license. The primary data set was retrieved from the website.

Hold the CTRL key while you click to download the following file:

AB_NYC_2019.csv

This data set has 16 fields and 48895 rows.

Attributes of this data set that have been used for this project are limited to the following:

Column	Type	Description
id	Number	Unique identifier for each Airbnb listing.
host_id	Number	Unique identifier for the host of the listing.
price	Number	Price of the listing per night.
minimum_nights	Number	Minimum number of nights required for a reservation.
number_of_reviews	Number	Total number of reviews received by the listing.
reviews_per_month	Number	Average number of reviews received by the listing per month.
calculated_host_listing_count	Number	Total number of listings hosted by the same host.
availability_365	Number	Number of days out of the year the listing is available for booking.
latitude	Number	Geographic coordinates of the listing's location.
longitude	Number	Geographic coordinates of the listing's location.
Name	String	Name of the listing
host_name	String	Name of the host
neighbourhood_group	String	The broader area surrounding the listing (e.g., Brooklyn, Manhattan).
neighbourhood	String	A specific neighborhood within the city (e.g., Williamsburg, SoHo).
room_type	String	Type of accommodation offered (e.g., Entire Place, Private Room, Shared Room).
last_review	String	Date of the most recent review received by the listing.

Tasks in the project:

Task 1. Building hierarchies

1. Create a hierarchy named *Neighbourhood Hierarchy* using **Neighbourhood** and **Neighbourhood Group**.
2. Create a separate hierarchy named *Room Type Hierarchy* for different Room Type, Entire home/apt, Private room, and Shared room. There is no separate field like *Entire home/apt*, *Private room* or *Shared room*. First, create a calculated field *Room Category* using the **If-then-else** command and then include that in *Room Type Hierarchy*.

Task 2. Simple data modeling

1. Create a calculated field named *Occupancy Rate* by dividing **Availability 365** by 365.
2. Filter the fields **Price**, **Occupancy Rate**, and **Minimum Nights** so that you can work non-null values for those fields. You are using Tableau Desktop [Public Edition] so use the **Filter** shelf for filtering.
3. Calculate a new field *Average Price per Night* by dividing **Price** by **Minimum Nights**.

Task 3. Creating basic visualizations

1. Create a Line Chart to visualize the relationship between **Price** and **Reviews Per Month**.
2. Create a Map using **Neighbourhood Group** and **Neighbourhood**.
3. Create a Bar Chart to compare the previously created field **Average Price per Night** across **Neighbourhood Group**.

Task 4. Create LOD and complex calculations (by calculated field)

1. Create a calculated field named *Price Difference* using Level of Detail (LOD) calculations to show the difference between a listing's **Price** and the average price in its neighborhood.
2. Create a calculated field named *Total Revenue* by multiplying **Price** by **Minimum Nights**.

Task 5. Using tooltips

1. Enhance your Map visualizations by adding tooltips that display additional information on hover, such as **Calculated Host Listings Count**, **Price Difference**, **Total Revenue**, and **Occupancy Rate**.

Task 6. Creating a final dashboard

1. Create a dashboard and arrange the created visualizations (Map and Bar chart) and calculated fields into a visually appealing and informative dashboard.
2. Use titles, labels, and annotations to guide viewers through the analysis. Titles, labels and annotations are not mentioned as you can show your creativity here.

Task 7. Implementing dashboard actions

1. Create a filter action that allows users to select a specific **Neighbourhood Group** on the Bar Chart and see the corresponding price distribution and other details in each neighborhood on the map.
2. Implement a highlight action that emphasizes listings of a specific **Neighbourhood Group** when selected on a chart.

Task 8. Project deliverables

1. Publish your final project workbook and copy the link to it for submission.

Submission

You will share the link to your Tableau workbook so your peers can assess your project. This project assesses you on the clarity of your presentation, the cohesiveness of your data storytelling, and the overall impact of your data visualizations. Follow the provided instructions and project rubric to showcase your Tableau skills effectively.

Author(s)

Shubhra Das